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25K-6544
BSCE-2A

Q2:

Code:

```
#include <stdio.h>
#include <string.h>
```

```
int main() {
    char password[] = "xyz123";
    char input[50];
    int attempts = 0;

    while (attempts < 3) {
        printf("Enter password: ");
        scanf("%s", input);

        if (strlen(input) == 0) {
            printf("Empty password!\n");
            continue;
        }

        if (strcmp(input, password) == 0) {
            printf("Access Granted!\n");
            return 0;
        } else {
            attempts++;

            if (attempts == 2) {
```

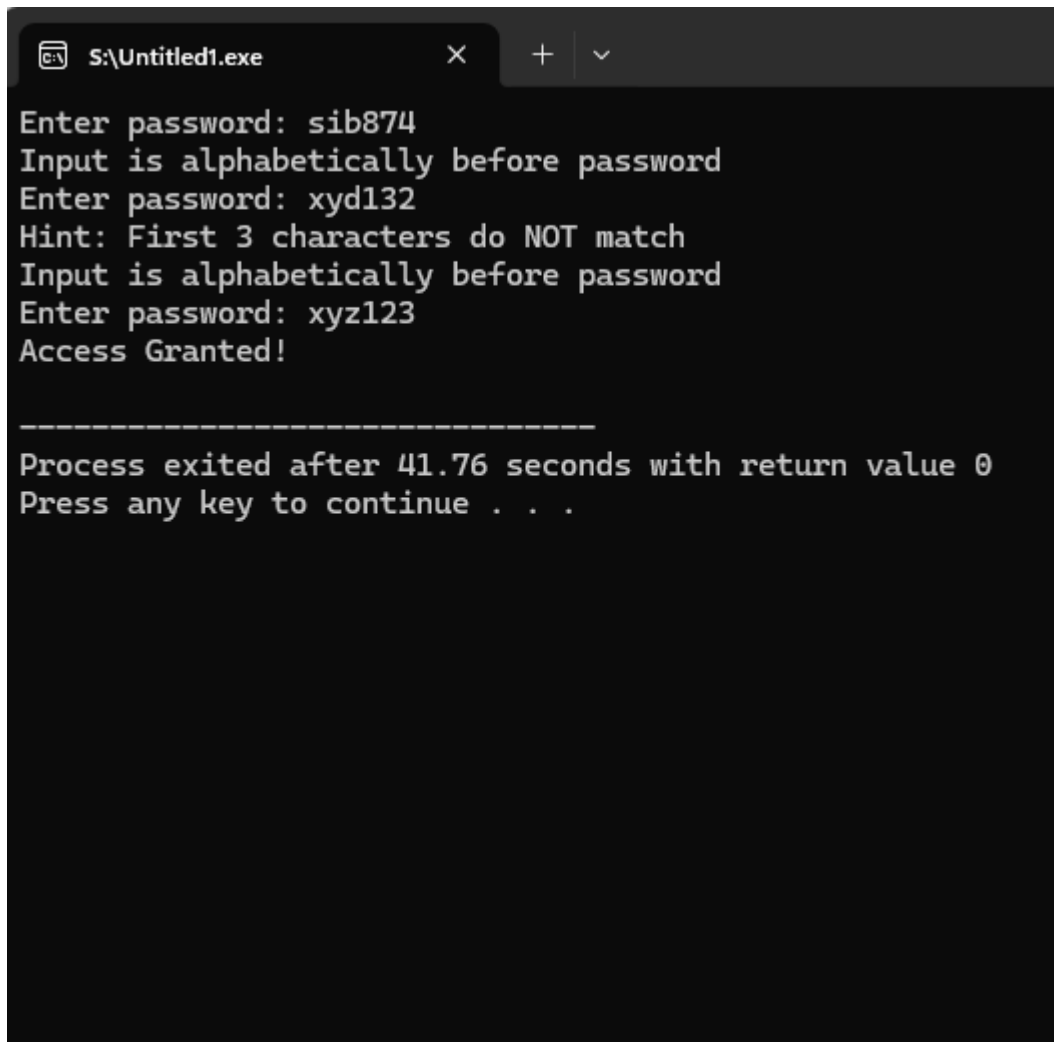
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```
        if (strncmp(input, password, 3) == 0)
            printf("Hint: First 3 characters match\n");
        else
            printf("Hint: First 3 characters do NOT match\n");
    }

    if (strcmp(input, password) < 0)
        printf("Input is alphabetically before password\n");
    else
        printf("Input is alphabetically after password\n");
}

printf("Account locked after 3 attempts\n");
return 0;
```

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```
S:\Untitled1.exe
Enter password: sib874
Input is alphabetically before password
Enter password: xyd132
Hint: First 3 characters do NOT match
Input is alphabetically before password
Enter password: xyz123
Access Granted!

-----
Process exited after 41.76 seconds with return value 0
Press any key to continue . . .
```

Q3:

Code:

```
#include <stdio.h>
#include <string.h>
```

```
int main() {
    char email[50], copy[50], format[100];
```

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```
printf("Enter email: ");
```

```
scanf("%s", email);
```

```
strcpy(copy, email);
```

```
char *ptr = strchr(copy, '@');
```

```
if (ptr == NULL) {
```

```
    printf("Invalid email!\n");
```

```
    return 0;
```

```
}
```

```
printf("Domain: %s\n", ptr + 1);
```

```
if (strstr(ptr + 1, ".") == NULL) {
```

```
    printf("Invalid domain!\n");
```

```
    return 0;
```

```
}
```

```
strcpy(format, "Email: ");
```

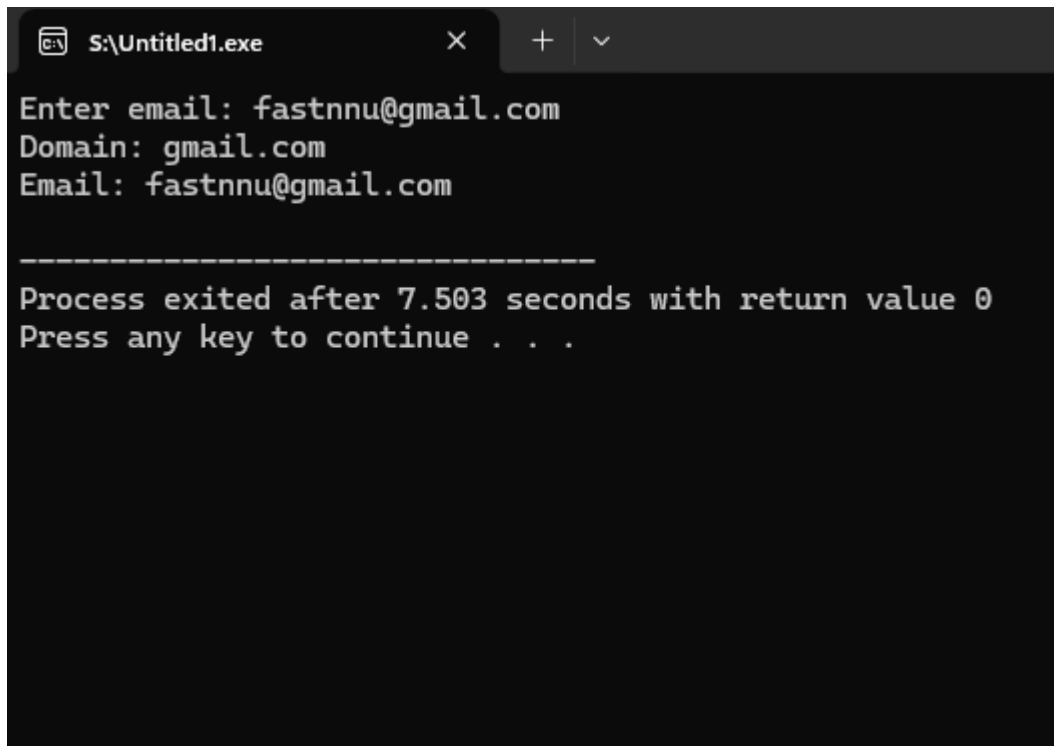
```
strcat(format, email);
```

```
printf("%s\n", format);
```

```
return 0;
```

```
}
```

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```
S:\Untitled1.exe
Enter email: fastnnu@gmail.com
Domain: gmail.com
Email: fastnnu@gmail.com

-----
Process exited after 7.503 seconds with return value 0
Press any key to continue . . .
```

Q4:

Code:

```
#include <string.h>
```

```
int main() {
    char names[5][30];
    char search[30];
    int x, found = 0;

    for (x = 0; x < 5; x++) {
        printf("Enter name %d: ", x + 1);
        fgets(names[x], 30, stdin);
        names[x][strcspn(names[x], "\n")] = '\0';
    }
}
```

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```
printf("\nStudent List:\n");  
for (x = 0; x < 5; x++)  
    printf("%s\n", names[x]);
```

```
printf("\nEnter name to search: ");  
fgets(search, 30, stdin);  
search[strcspn(search, "\n")] = '\0';
```

```
for (x = 0; x < 5; x++) {  
    if (strcmp(names[x], search) == 0) {  
        printf("Found at position %d\n", x);  
        found = 1;  
        break;  
    }  
}
```

```
if (!found)  
    printf("Student not found\n");
```

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```
S:\Untitled1.exe X + v
Enter name 1: sibtain
Enter name 2: shaikh
Enter name 3: ibaad
Enter name 4: huzaifa
Enter name 5: saim

Student List:
sibtain
shaikh
ibaad
huzaifa
saim

Enter name to search: saim
Found at position 4

-----
Process exited after 32.62 seconds with return value 0
Press any key to continue . . .
```

Q5:

Code:

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main() {
    char words[6][20];
    int seen[6] = {0};
    int i, j;
```

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```
for (i = 0; i < 6; i++) {  
    printf("Enter word %d: ", i + 1);  
    scanf("%s", words[i]);  
}
```

```
for (i = 0; i < 6; i++) {  
    if (seen[i] == 1)  
        continue;
```

```
    int count = 1;
```

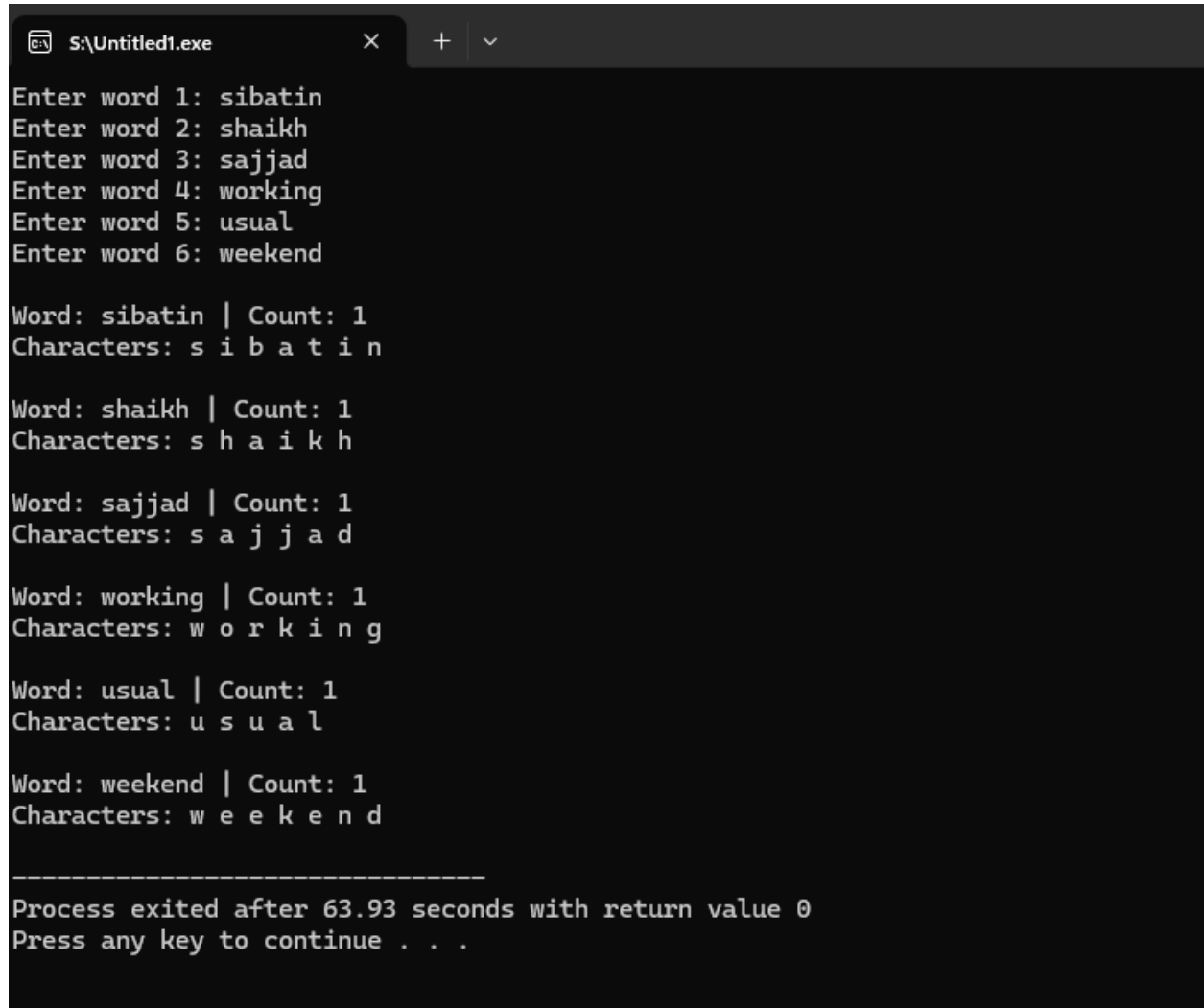
```
    for (j = i + 1; j < 6; j++) {  
        if (strcmp(words[i], words[j]) == 0) {  
            count++;  
            seen[j] = 1;  
        }  
    }  
}
```

```
printf("\nWord: %s | Count: %d\n", words[i], count);
```

```
printf("Characters: ");  
for (j = 0; words[i][j] != '\0'; j++) {  
    printf("%c ", words[i][j]);  
}  
printf("\n");  
}
```


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return 0



```
S:\Untitled1.exe
Enter word 1: sibatin
Enter word 2: shaikh
Enter word 3: sajjad
Enter word 4: working
Enter word 5: usual
Enter word 6: weekend

Word: sibatin | Count: 1
Characters: s i b a t i n

Word: shaikh | Count: 1
Characters: s h a i k h

Word: sajjad | Count: 1
Characters: s a j j a d

Word: working | Count: 1
Characters: w o r k i n g

Word: usual | Count: 1
Characters: u s u a l

Word: weekend | Count: 1
Characters: w e e k e n d

-----
Process exited after 63.93 seconds with return value 0
Press any key to continue . . .
```

Q6:

Code:

```
int main() {  
    FILE *fptr;
```

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```
char name[50];
```

```
int grade, x;
```

```
fptr = fopen("grades.txt", "w");
```

```
if (fptr == NULL) {
```

```
    printf("File error!\n");
```

```
    return 0;
```

```
}
```

```
for (x = 0; x < 3; x++) {
```

```
    printf("Enter name and grade: ");
```

```
    scanf("%s %d", name, &grade);
```

```
    fprintf(fptr, "%s %d\n", name, grade);
```

```
}
```

```
fclose(fptr);
```

```
fptr = fopen("grades.txt", "r");
```

```
printf("\nRecords:\n");
```

```
while (fscanf(fptr, "%s %d", name, &grade) != EOF) {
```

```
    printf("%s %d\n", name, grade);
```

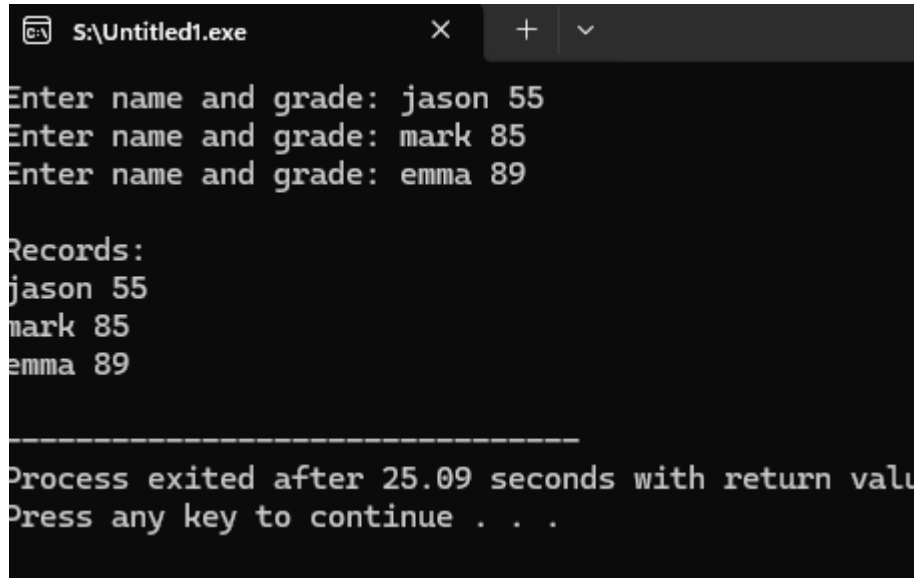
```
}
```

```
fclose(fptr);
```

```
return 0;
```

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}



```
S:\Untitled1.exe
Enter name and grade: jason 55
Enter name and grade: mark 85
Enter name and grade: emma 89

Records:
jason 55
mark 85
emma 89

-----
Process exited after 25.09 seconds with return value 0
Press any key to continue . . .
```

Q7:

Code:

```
#include <stdio.h>
```

```
int main() {
```

```
    FILE *fptr;
```

```
    char product[30];
```

```
    int qty;
```

```
    fptr = fopen("inventory.txt", "a");
```

```
    if (fptr == NULL) {
```

```
        printf("File error!\n");
```

```
        return 0;
```

```
    }
```

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```
printf("Enter product and quantity: ");
```

```
scanf("%s %d", product, &qty);
```

```
fprintf(fp, "%s %d\n", product, qty);
```

```
fclose(fp);
```

```
fp = fopen("inventory.txt", "r");
```

```
printf("\nInventory:\n");
```

```
while (fscanf(fp, "%s %d", product, &qty) != EOF) {
```

```
    printf("%s %d\n", product, qty);
```

```
}
```

```
fclose(fp);
```

```
return 0;
```

```
}
```

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```
S:\Untitled1.exe X + v
Enter product and quantity: rice 44

Inventory:
flour 23
flour 80
sugar 34
rice 44

-----
Process exited after 4.64 seconds with return value 0
Press any key to continue . . .
```

FOR 4 OUTPUTS, RUNTIME =4;

Q8:

CODE:

```
int main() {
    FILE *fptr;
    int i, score, sum = 0, val;

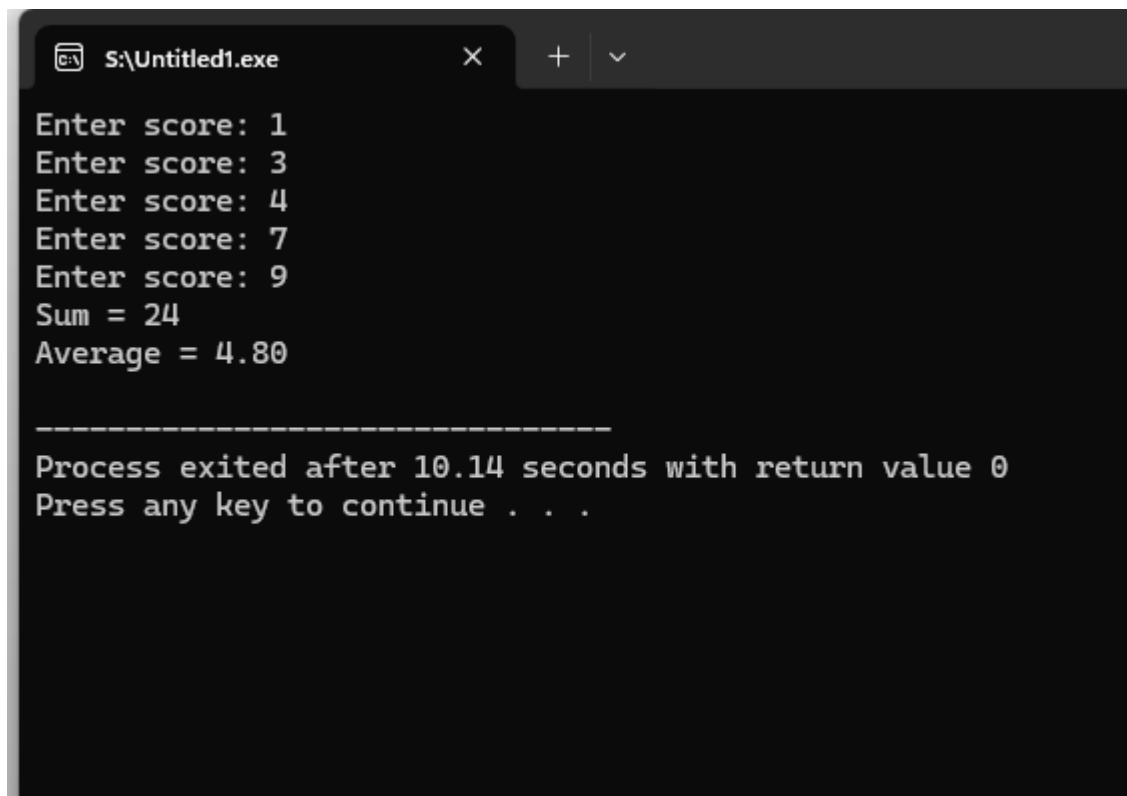
    fptr = fopen("survey.txt", "w");

    for (i = 0; i < 5; i++) {
```

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```
    printf("Enter score: ");  
    scanf("%d", &score);  
    fprintf(fptr, "%d\n", score);  
}  
  
fclose(fptr);  
  
fptr = fopen("survey.txt", "r");  
  
while (fscanf(fptr, "%d", &val) != EOF) {  
    sum += val;  
}  
  
float avg = sum / 5.0;  
  
printf("Sum = %d\n", sum);  
printf("Average = %.2f\n", avg);  
  
fclose(fptr);  
return 0
```

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```
S:\Untitled1.exe
Enter score: 1
Enter score: 3
Enter score: 4
Enter score: 7
Enter score: 9
Sum = 24
Average = 4.80

-----
Process exited after 10.14 seconds with return value 0
Press any key to continue . . .
```

Q9;

CODE:

```
#include <stdio.h>
```

```
int main() {
```

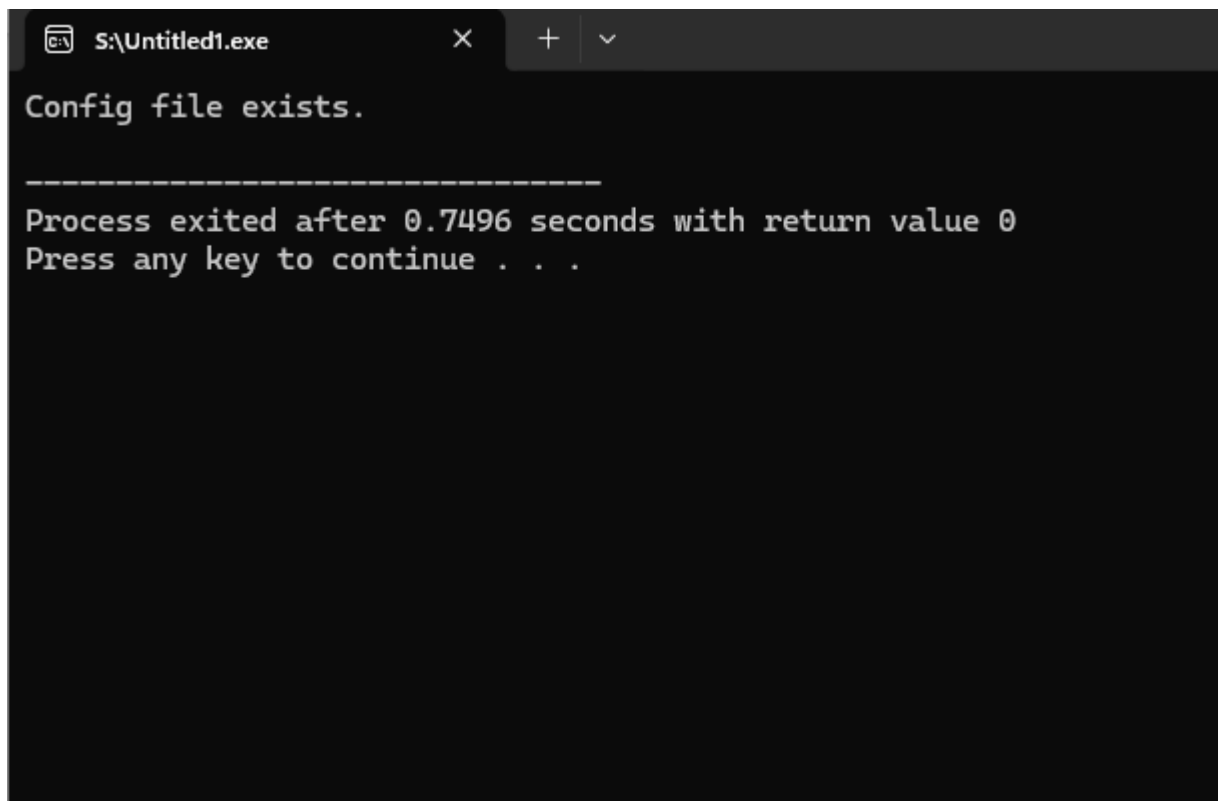
```
    FILE *fptr;
```

```
    fptr = fopen("config.txt", "r");
```

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```
if (fptr == NULL) {  
    printf("File not found. Creating default config...\n");  
  
    fptr = fopen("config.txt", "w");  
  
    fprintf(fptr, "max_users=50\n");  
    fprintf(fptr, "timeout=30\n");  
    fprintf(fptr, "mode=default\n");  
  
    fclose(fptr);  
} else {  
    printf("Config file exists.\n");  
    fclose(fptr);  
}  
  
return 0;  
}
```


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```
S:\Untitled1.exe X + v
Config file exists.
-----
Process exited after 0.7496 seconds with return value 0
Press any key to continue . . .
```

Q10:

CODE:

```
#include <stdio.h>
```

```
int main() {
    FILE *fptr;
    char name[30];
    int s1, s2, s3;
    float avg;
```

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```
printf("Enter name: ");
```

```
scanf("%s", name);
```

```
printf("Enter 3 scores: ");
```

```
scanf("%d %d %d", &s1, &s2, &s3);
```

```
avg = (s1 + s2 + s3) / 3.0;
```

```
fptr = fopen("report.txt", "w+");
```

```
fprintf(fptr, "Name: %s\n", name);
```

```
fprintf(fptr, "Scores: %d %d %d\n", s1, s2, s3);
```

```
fprintf(fptr, "Average: %.2f\n", avg);
```

```
if (avg >= 50)
```

```
    fprintf(fptr, "Status: Pass\n");
```

```
else
```

```
    fprintf(fptr, "Status: Fail\n");
```

```
rewind(fptr);
```

```
char line[100];
```

```
printf("\nReport Card:\n");
```

```
while (fgets(line, sizeof(line), fptr)) {
```

```
    printf("%s", line);
```

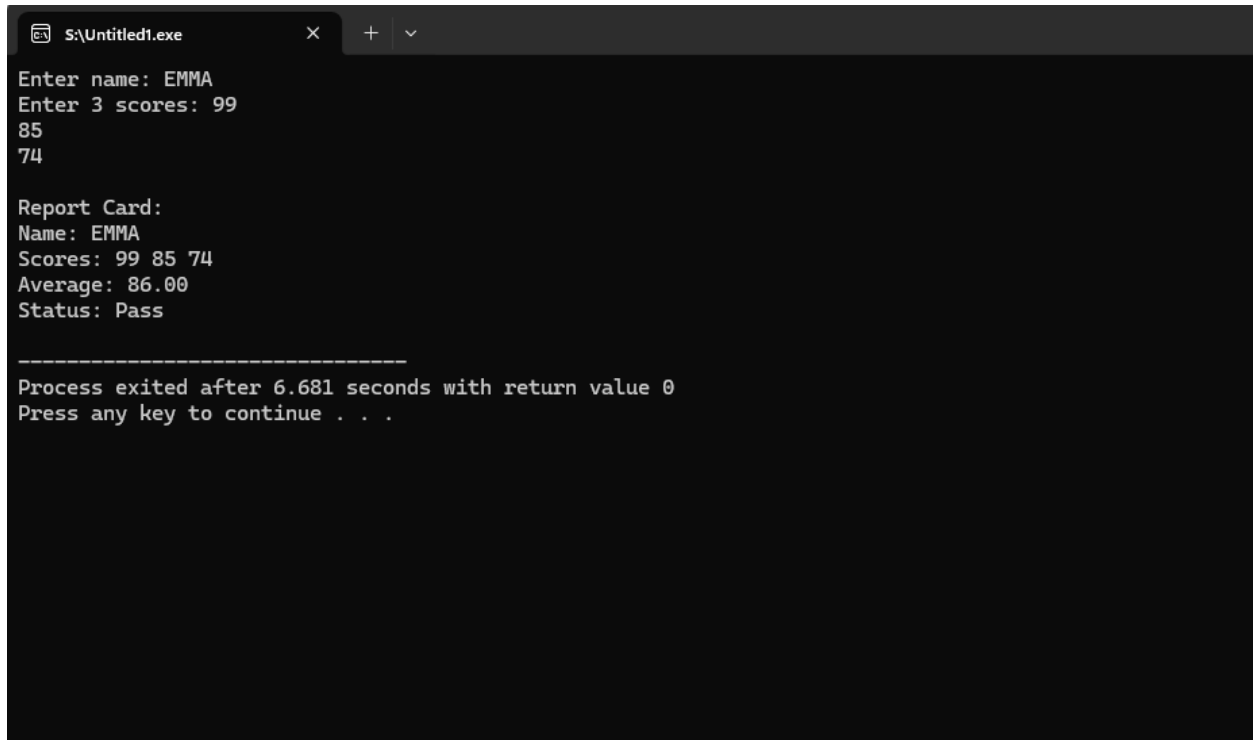
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```
}
```

```
fclose(fptr);
```

```
return 0;
```

```
}
```



```
S:\Untitled1.exe
Enter name: EMMA
Enter 3 scores: 99
85
74

Report Card:
Name: EMMA
Scores: 99 85 74
Average: 86.00
Status: Pass

-----
Process exited after 6.681 seconds with return value 0
Press any key to continue . . .
```